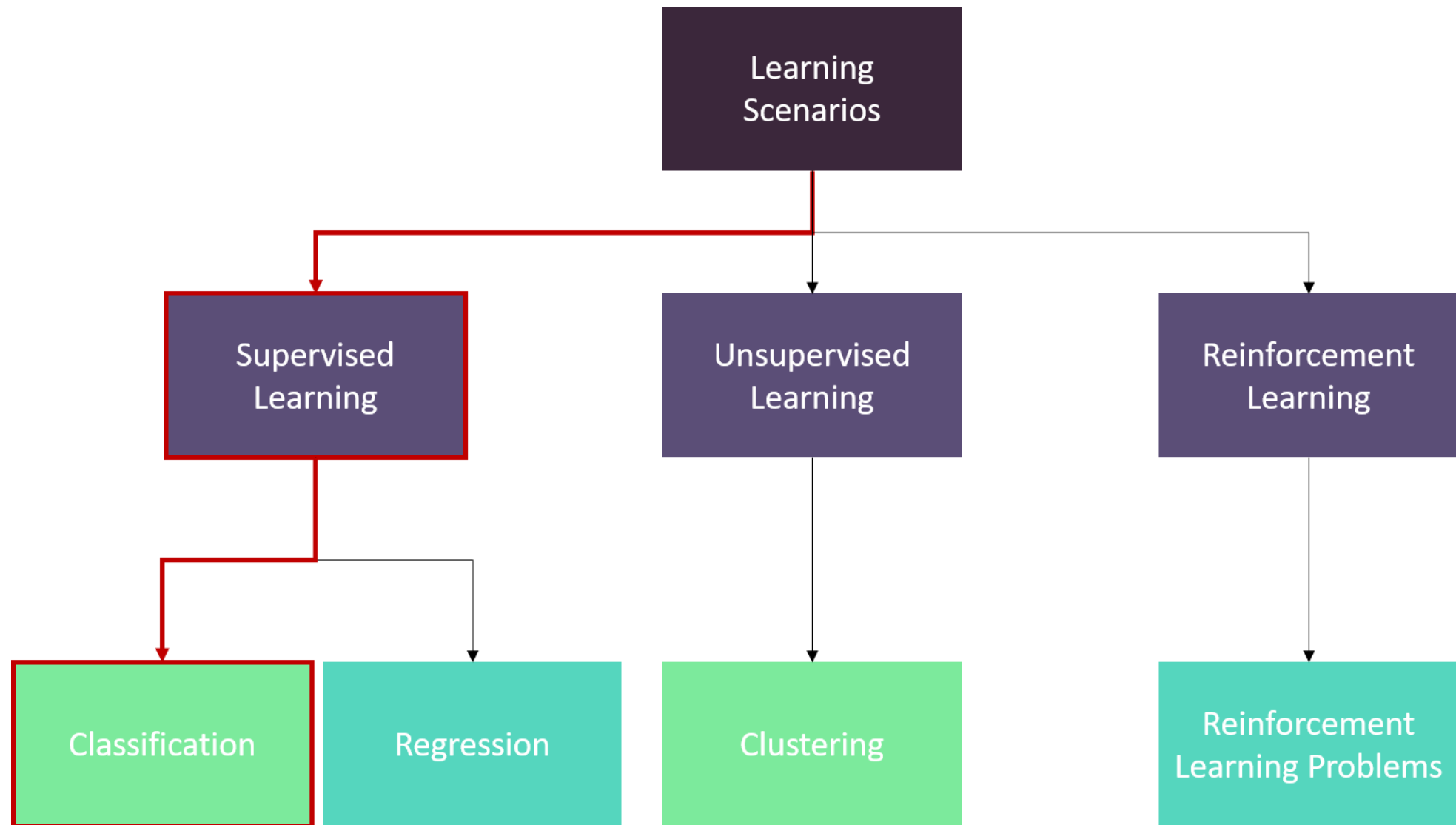
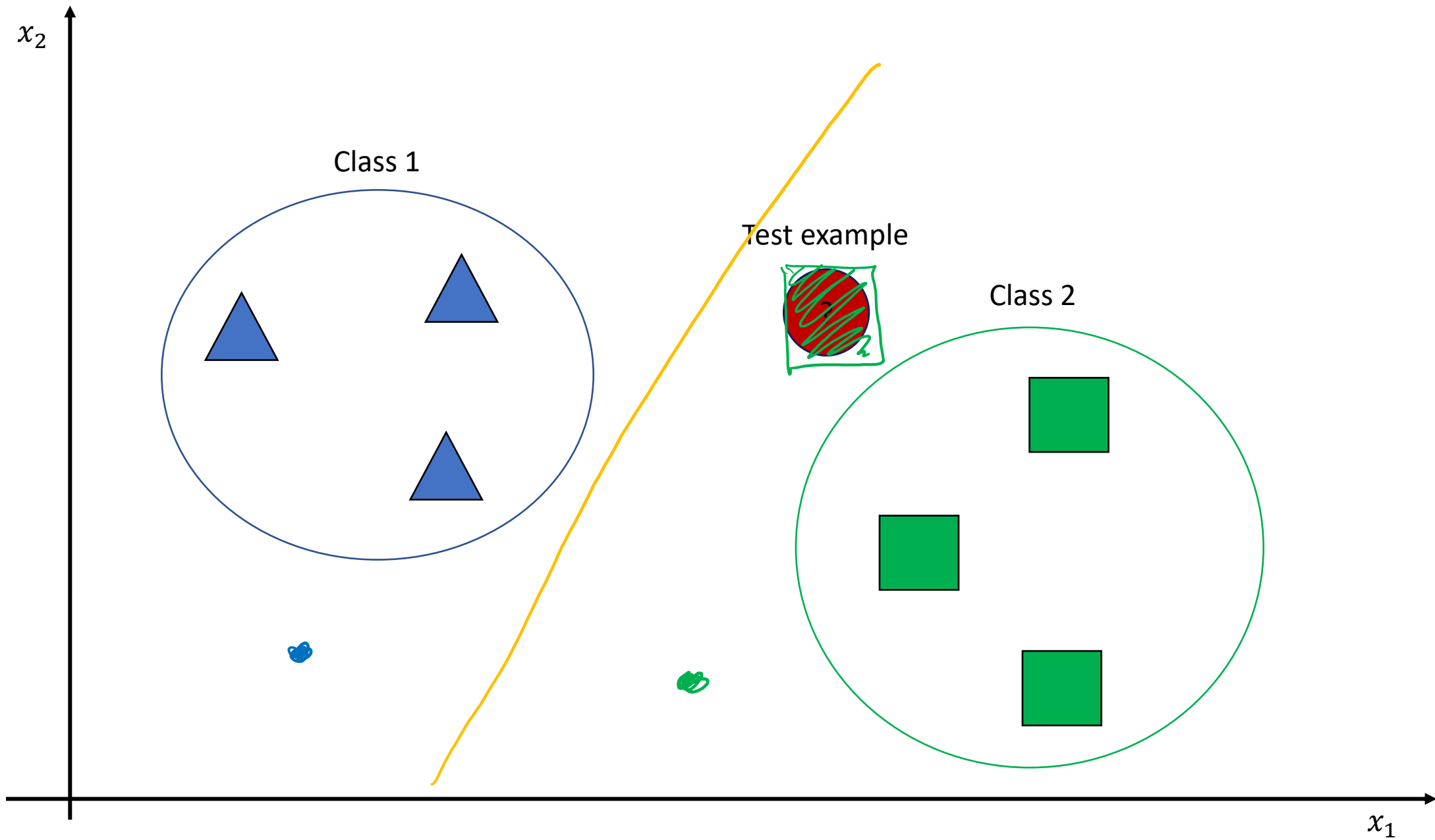


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Nearest Neighbour Algorithms





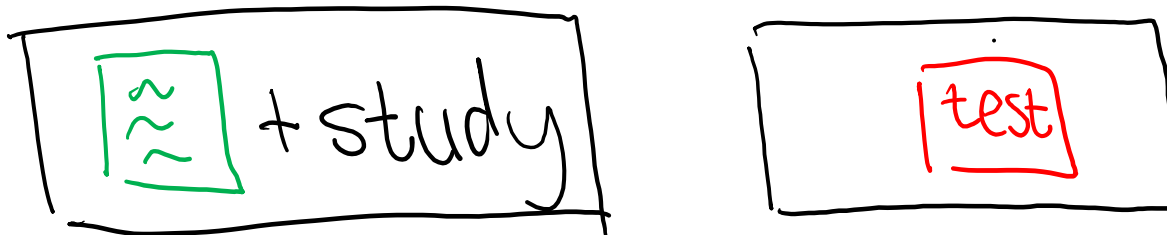


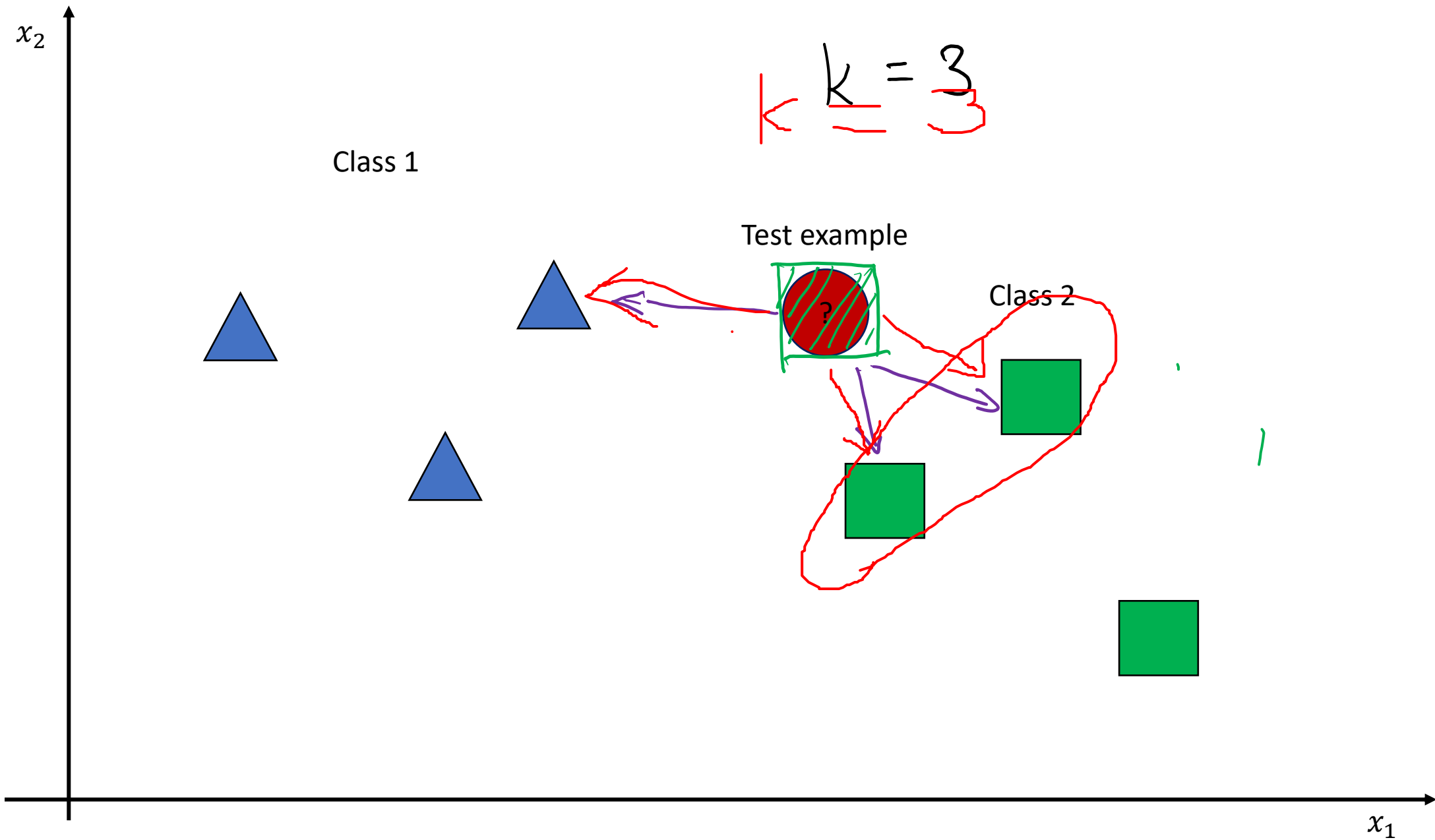
Nearest Neighbour Algorithm

- *Lazy* approach to learning
- Training data used during testing
 - Similar to an open-book test



- Compare to *eager* learners that learn prior to testing
 - Training data not used during testing





The scenario

- Training set $\nearrow f(x^{[i]})$
 - $D = \{(\mathbf{x}^{[i]}, \mathbf{y}^{[i]}), \dots, (\mathbf{x}^{[n]}, \mathbf{y}^{[n]})\}$

- K-Nearest Neighbours $\leftarrow \mathbb{R}$
 - $D_k = \{(\mathbf{x}^{[i]}, \mathbf{y}^{[i]}), \dots, (\mathbf{x}^{[k]}, \mathbf{y}^{[k]})\}$

- Classification
 - $h(\mathbf{x}^{[t]}) = \text{mode}(\{\mathbf{y}^{[i]}, \dots, \mathbf{y}^{[k]}\})$

- Regression $\text{mode}(\{1, 1, 2, 3\})$
 - $h(\mathbf{x}^{[t]}) = \frac{1}{k} \sum_{i=1}^k \mathbf{y}^{[i]}$

$$f: \mathbb{R}^m \rightarrow \mathbb{R}$$

$\downarrow$
 $y^1 = 0.9, y^2 = 0.7$

$$f(x) = y$$

classes $\nearrow$

$$y \in \{1, \dots, t\}$$

$$f: \mathbb{R}^{\boxed{m}} \rightarrow \{1, \dots, t\}$$

$\rightarrow$ # features

$$d(x^{[t]}, x^{[i]}) =$$

$$\sqrt{\sum_{j=1}^m (x_j^{[t]} - x_j^{[i]})^2}$$

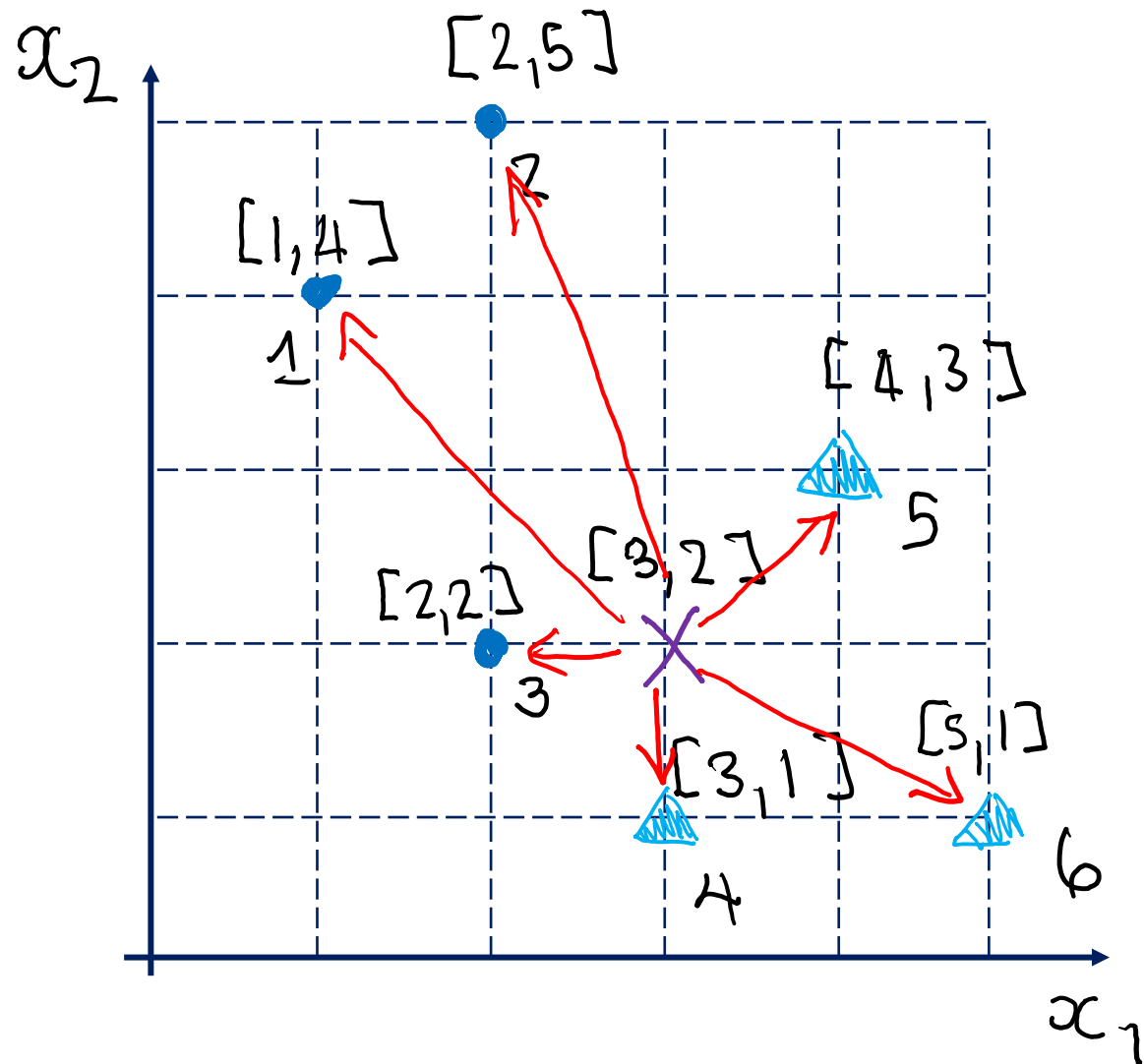


Pros and cons

- Easy to implement
- No assumptions about data – *non-parametric*
- Limited by memory constraints
- Computational complexity
 - $O(n \times m)$
 - n = # training examples
 - m = # of features
 - Assuming $n \gg m$, $O(n)$
- Curse of Dimensionality



Numerical Example



$$\mathcal{D} = \{(\underbrace{[1, 4]}_{X^{[1]}}, \underbrace{1}_{Y^{[1]}}, \dots, \underbrace{[5, 1]}_{X^{[6]}}, \underbrace{0}_{Y^{[6]}})\}$$

$$X^{[t]} = [3, 2]$$

$$d(\underbrace{X^{[t]}}_1, \underbrace{X^{[i]}}_i) = \sqrt{\sum_{j=1}^m (\underbrace{x_j^{[t]}}_2 - \underbrace{x_j^{[i]}}_3)^2}$$

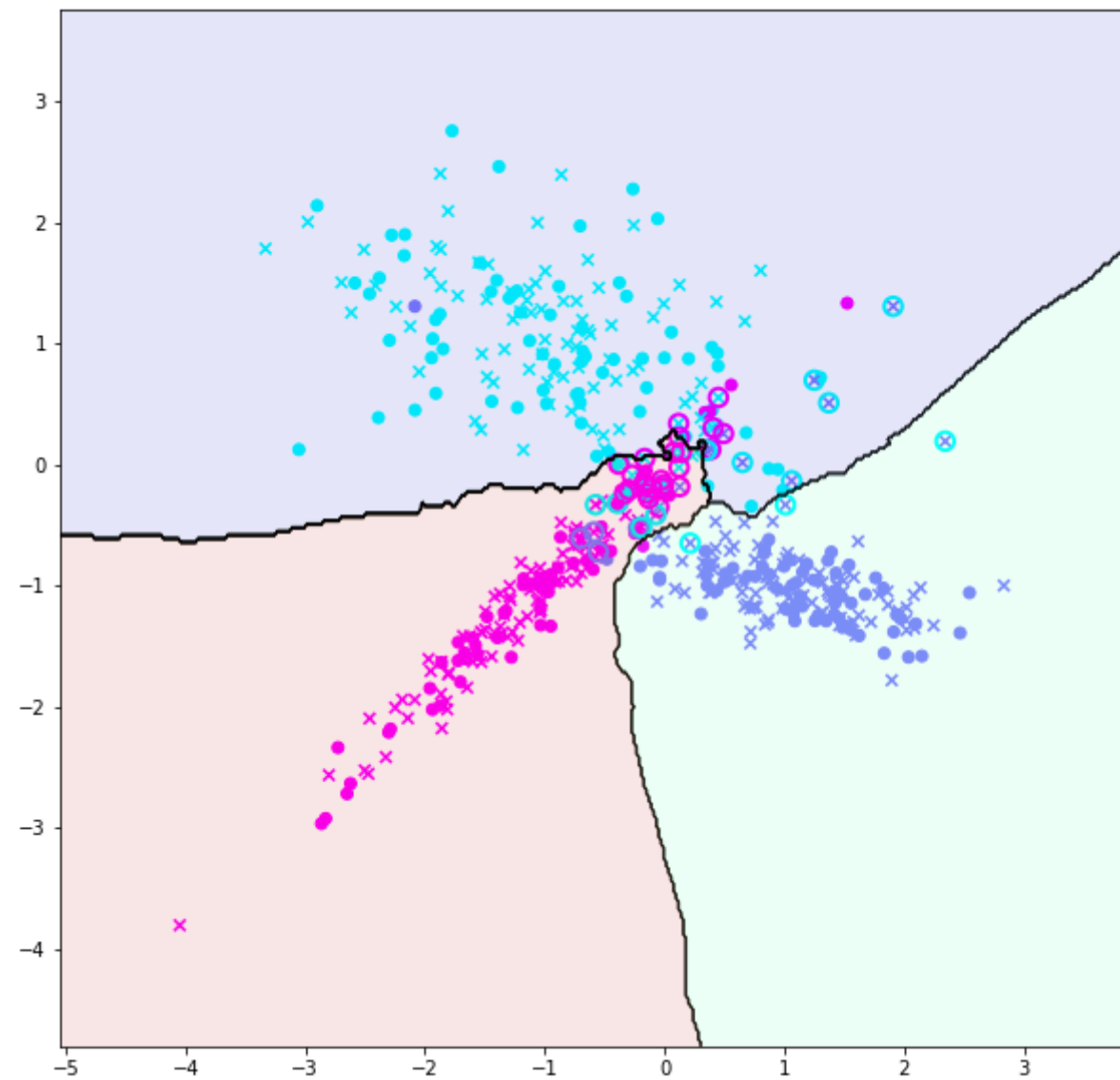
$$[2.83, 3.16, 1.0, 1.0, 1.41, 2.23]$$

$$\text{argsort} = [3, 4, 5, \mid 6, 1, 2]$$

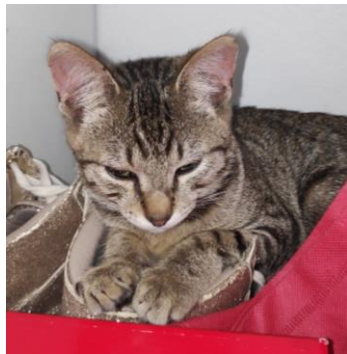
$$h(X^{[t]}) = 0$$

$$y[3, 4, 5] = \{1, 0, 0\} \quad k=3 \rightarrow$$

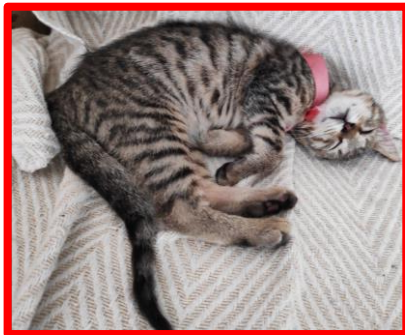




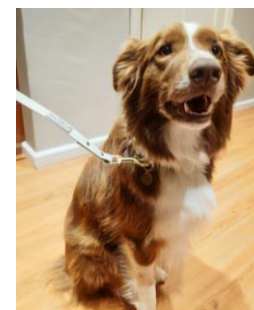
x_2



Juniper



Cooper



x_1



Next up:

kNN Examples

